

Name = Sadashiv

USN = 2VX24UE044

Expt = 3

Title = Merge sort

```
In [1]: import random
import time
import matplotlib.pyplot as plt

def merge_sort(arr):
    if len(arr) <= 1:
        return arr

    mid = len(arr)//2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])
    return merge(left, right)

def merge(left, right):
    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    return result + left[i:] + right[j:]

arr = list(map(int, input("Enter elements separated by space:").split()))
print("Input array:", arr)

start_time = time.time()
sorted_arr = merge_sort(arr)
end_time = time.time()

print("Sorted array:", sorted_arr)
print("Time taken to sort:", end_time - start_time, "seconds")
print("=====")

n_values = [5000, 6000, 7000, 8000]
time_values = []

for n in n_values:
    arr = [random.randint(1, 9) for _ in range(n)]

    start_time = time.time()
    merge_sort(arr)
    end_time = time.time()

    time_taken = end_time - start_time
    time_values.append(time_taken)
    print("Time for", n, "elements:", time_taken)
print("=====")

plt.plot(n_values, time_values, color = 'red', marker='o')
plt.xlabel('Number of Elements (n)')
plt.ylabel('Time Taken (seconds)')
plt.title('Qui Sort Time Complexity Analysis')
plt.grid(True)
plt.show()
```

Enter elements separated by space:20 50 30 40 10 90

Input array: [20, 50, 30, 40, 10, 90]

Sorted array: [10, 20, 30, 40, 50, 90]

Time taken to sort: 0.0 seconds

=====

Time for 5000 elements: 0.021593809127807617

Time for 6000 elements: 0.03760814666748047

Time for 7000 elements: 0.030286788940429688

Time for 8000 elements: 0.030335187911987305

=====

